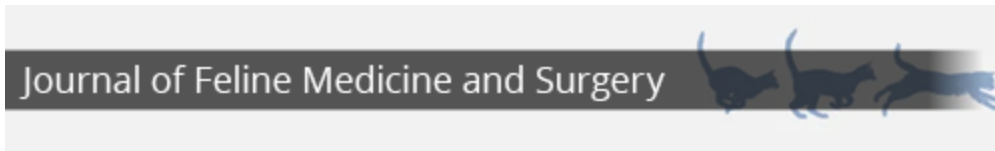


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Therapies for Feline Chronic Kidney Disease

What is the Evidence?

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Abstract



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Practical relevance Successful treatment and prevention of kidney disease in pet animals requires a multidimensional approach to identify and eliminate causes or exacerbating factors, provide professional examination and care on a regular basis, and plan and implement a comprehensive treatment program when necessary.

Evidence base Over the years, many therapeutic and preventive interventions have been developed or advocated for chronic kidney disease (CKD), but evidence of efficacy or effectiveness is often lacking or highly variable. Accordingly, the main objective of this systematic review was to identify and critically appraise the evidence supporting various approaches to managing feline CKD; namely, fluid therapy, calcitriol therapy, antihypertensive therapy, ACE inhibitor therapy, erythropoietic hormone replacement therapy, potassium supplementation, antioxidant supplementation, alkalization therapy, dietary phosphorus restriction and intestinal phosphate binders, therapeutic renal foods, assisted feeding, dialysis and renal transplantation.

Chronic kidney disease defined and staged

It has recently been proposed that the term ‘renal failure’ be discarded in favor of ‘kidney disease’ for two reasons. First, pet owners know what a kidney is, but many are unfamiliar with the medical term ‘renal.’ Thus using the term ‘kidney’ facilitates communication between veterinarians and clients. Secondly, there has never been complete agreement on the specific meanings of renal insufficiency and renal failure. By using the term ‘kidney disease’ and staging the severity of disease, it is possible to greatly facilitate understanding, communication and application of management guidelines for patients.

Kidney disease is defined as the presence of functional or structural abnormalities in one or both kidneys.¹ It may be recognised by reduced kidney function or the presence of kidney damage. Kidney damage is defined as either: microscopic or macroscopic pathology detected by biopsy or direct visualization of the kidneys; or markers of damage detected by blood or urine tests or imaging studies.

The severity of kidney disease varies greatly depending on the magnitude of kidney involvement.

Patients with chronic kidney disease (CKD, as defined below) can be categorized into stages along a continuum of progressive disease.³ The International Renal Interest Society (IRIS) has proposed a four-tier system for staging CKD in dogs and cats (see box, page 196).⁴ The staging system was developed by a panel of experts in veterinary nephrology, rather than specific clinical studies, and its use has been advocated by the American and European Societies of Veterinary Nephrology and

Urology. The stage is based on the level of kidney function signified by at least two sequential determinations of serum creatinine concentration obtained in a well hydrated patient fasted for 12–24 h. Ideally, serum creatinine concentrations should be determined at least twice over 1–2 weeks to stage CKD. As a variety of factors affect serum creatinine concentrations, published reference ranges are often exceedingly broad. Using the IRIS staging system, some patients classified as having mild renal azotemia (stage 2) may have serum creatinine values within published reference ranges. As a consequence, the patient's overall clinical status should be considered when interpreting serum creatinine concentration and other laboratory tests and when planning patient management. In particular, the serum creatinine concentration must be interpreted in the light of a concurrently obtained measure of urine concentrating ability (urine specific gravity or osmolality). In the absence of urinary obstruction, serum creatinine values > 1.6 mg/dl ($140\text{ }\mu\text{mol/l}$) are likely to indicate kidney disease when the patient's urine specific gravity is < 1.035 . Daily production of creatinine in the body is determined largely by the muscle mass of the individual, so care must be taken in interpreting values obtained from animals with reduced muscle mass; for example, elderly cats with CKD and poor muscle mass may have lower than expected serum creatinine concentrations. [5](#)

CKD is defined as kidney damage present for at least 3 months, with or without decreased glomerular filtration rate (GFR), or a $> 50\%$ reduction in GFR persisting for at least 3 months. [2](#)

Stages of feline CKD

Stage 1 Non-azotemia

Markers of kidney damage present

Plasma creatinine < 140 $\mu\text{mol/l}$ (1.6 mg/dl)

Proteinuria: classify (P/NP/BP)*

Hypertension: classify (N/L/M/H/nc/c/RND)†

Stage 2 Mild renal azotemia

Markers of kidney damage present

Plasma creatinine 140–249 $\mu\text{mol/l}$ (1.6–2.8 mg/dl)

Proteinuria: classify (P/NP/BP)*

Hypertension: classify (N/L/M/H/nc/c/RND)†

stage 3 Moderate renal azotemia

Plasma creatinine 250–439 $\mu\text{mol/l}$ (2.9–5.0 mg/dl)

Proteinuria: classify (P/NP/BP)*

Hypertension: classify (N/L/M/H/nc/c/RND)†

Stage 4 Severe renal azotemia

Plasma creatinine $\geq 440 \mu\text{mol/l}$ (5.0 mg/dl)

Proteinuria: classify (P/NP/BP)*

Hypertension: classify (N/LyM/H/nc/c/RND)†

For full details of the staging system, see the International Renal Interest Society's website:
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P = proteinuria (protein:creatinine ratio [UPC] > 0.4), NP = non-proteinuria (UPC < 0.2), BP
= borderline proteinuria (UPC 0.2–0.4]

N = minimal risk of complications, L = low risk of complications, M = moderate risk of complications, H = high risk of complications, nc = no complications, c = complications, RND = risk not determined (blood pressure not measured)

The IRIS has proposed to further subclassify patients with CKD according to their magnitude of proteinuria and systemic blood pressure. Proteinuria and hypertension appear to influence prognosis; however, the impact of therapeutic intervention to lessen proteinuria and hypertension on long-term prognosis is still uncertain.⁶⁻⁹ Classification of patients as proteinuric necessitates determining the urine protein:creatinine ratio (UPC) and eliminating the presence of hemorrhage, inflammation and/or infection. Feline patients are classified as proteinuric when their UPC > 0.4. Values for UPC between 0.2 and 0.4 are classified as borderline proteinuric, while UPC values < 0.2 are classified as non-proteinuric. Ideally, UPC values between 0.4 and 1.0 should be confirmed using several urine samples obtained over 2–4 weeks. With respect to blood pressure, the IRIS has proposed that patients be further staged according to the degree of risk of end-organ damage and whether there is current evidence of such damage or complications.⁴

Goals and limits of therapy for CKD

Conservative medical management of CKD consists of supportive and symptomatic therapy designed to correct abnormalities in fluid, electrolyte, acid-base, endocrine and nutritional balance. Therapy aims to minimize the clinical and pathophysiological consequences of reduced kidney function. In general, this type of management should not be expected to halt, reverse or eliminate renal lesions responsible for CKD. Therefore, management strategies are most beneficial when combined with specific therapy directed at correcting the primary cause of the kidney disease such as hypercalcemic nephropathy, bacterial urinary tract infections and obstructive uropathy. Efforts should be directed at detecting and treating active primary kidney diseases in addition to managing the various pathophysiological and clinical consequences of CKD.

Medical management guidelines provided here are intended for pets with compensated disease; not for patients unable to eat or accept oral medications because of severe uremia. Clinical signs of uremia must be corrected before instituting the general principles of medical management. The goals of medical management of CKD are to:

- ameliorate clinical signs of uremia;

minimize disturbances associated with excesses or losses of electrolytes, vitamins and minerals;
support adequate nutrition by supplying daily protein, calorie, mineral and other nutrient requirements; and
modify progression of kidney disease.²

Although evidence-based medicine does not always lead to a definitive answer, it does provide a framework for making decisions and understanding the risk-benefit relationship of various therapeutic or preventive plans.

These goals are best achieved when recommendations are individualized to the patient's needs on the basis of clinical and laboratory findings. Because CKD is progressive and dynamic, serial assessment of the patient and modification of therapy in response to changes are essential parts of the management strategy.

Evidence-based veterinary nephrology concepts

Evidence-based medicine (EBM) represents a major, although largely untested, intellectual advance when making clinical decisions and determining patient care.¹⁰⁻¹² EBM has been defined as the integration of the best research evidence with clinical expertise and patient values.¹⁰ Best research evidence means clinically relevant research, especially from patient-centered clinical studies. Clinical expertise refers to the use of clinical skills to identify each patient's unique health status, establish a diagnosis, and determine the risks and benefits of potential interventions. For veterinary medicine, the concept of patient values must include the unique preferences and expectations of each owner. In addition, if a therapeutic intervention is not readily available, then it is unlikely to benefit the patient. The basic tenet of EBM is that integration of these elements (clinically relevant research, clinical expertise, patient/owner preferences, and availability of resources) will result in the formation of a diagnostic and therapeutic alliance that optimizes clinical outcomes and quality of life.¹⁰

In the past, information based on pathophysiological rationale and data from models of kidney disease were often used to justify clinical recommendations in cats with CKD. The proliferation of randomized controlled clinical trials has led to an increase in the quantity and quality of clinically valid evidence. When possible, veterinarians should use information derived from systematically conducted, rigorously controlled clinical studies to make diagnostic and treatment decisions. Of course, not all recommendations can be based on such studies so it is important to recognize the inherent limitations of less rigorous forms of evidence. One method of accommodating concerns

regarding these limitations is to assign a score defining the strength and quality of the recommendation. The classification scheme proposed for veterinary clinical nutrition [13](#) may be useful for establishing rules of evidence for recommendations regarding veterinary nephrology. These guidelines categorize the quality of evidence into grades I to IV based on applicability to clinical case management (see box below). Grades I and II are evidence with the highest quality for application in the clinical environment, whereas grade IV would be evidence with the lowest quality.

The quality and strength of evidence can be used to make a recommendation about use of a specific therapeutic modality. Although EBM does not always lead to a definitive answer, it does provide a framework for making decisions and understanding the risk-benefit relationship of various therapeutic or preventive plans.

Quality of evidence grading guidelines

Grade I

Evidence obtained from one or more properly designed, randomized controlled clinical trials performed in clinical patients of the target species

Grade II

Evidence obtained from properly designed, randomized controlled studies performed using animals of the target species with spontaneous disease in a laboratory or research animal colony setting

Grade III

Evidence obtained from appropriately controlled studies without randomization, appropriately designed cohort or case control studies, studies using acceptable models of disease or simulations in the target species, case series or dramatic results from uncontrolled studies

Grade IV

Evidence obtained from studies conducted in other species, reports of expert committees, descriptive studies, case reports, pathophysiological justification, and opinions of recognized experts developed on the basis of their clinical experience

Applying evidence-based concepts to therapeutic decisions for CKD

A grade I to IV scoring system [13](#) was used to systematically evaluate the quality of published evidence for the use of 13 categories of interventions in cats with CKD, as discussed below. The strength of evidence was also assessed in making a final recommendation for use of a specific treatment modality.

It is important to note that the use of any specific therapeutic strategy hinges not only on the quality and strength of evidence summarized here but also on the clinical expertise of the individual veterinarian, owner preferences and available resources. In addition, the impact of providing unnecessary or unproven treatments on the pet-owner relationship should be considered. Treatments that are undesirable to the pet or owner may be disruptive to that relationship. As providing high quality health care to pets is based on the pet-family bond, it is important to avoid inadvertently disrupting this relationship when recommending therapy.

Fluid therapy

Progression of CKD is accompanied by decreases in GFR, worsening azotemia and ultimately uremia. Long-term administration of subcutaneous balanced electrolyte solutions has been advocated in patients with CKD to prevent dehydration, thereby maximizing renal blood flow and GFR, increasing urine output and ameliorating clinical manifestations of dehydration and uremia.^{2,14,15} Subcutaneous administration of balanced electrolyte solutions is usually reserved for cats with advanced (stages 3–4) CKD prone to dehydration.

Preliminary studies have been completed in healthy adult cats using intravenous fluids and various diuretics.¹⁶ In healthy cats, the renal effects of mannitol were superior to those of furosemide and dopamine together when used as an adjunct to fluid diuretic therapy.¹⁶ Additional studies with fluid therapy are required to assess physiological and clinical parameters in animals with CKD. Use of diuretics should be limited to patients with uremic crises rather than cats with stable CKD. Anecdotal reports suggest that many cat owners can give subcutaneous fluids successfully in the home environment for a long period of time. Hypertension is a well-recognized complication of CKD in cats and it is unknown if excessive sodium chloride delivered by parenteral fluids will exacerbate clinical problems associated with hypertension or kidney disease. For example, every 100 ml of lactated Ringer's solution provides 13 mEq sodium, which is approximately eight times the 24 h maintenance requirement for sodium in healthy or diseased cats.^{17,18} Although empirical evidence suggests that some individual cats may benefit from long-term subcutaneous fluid therapy, no controlled clinical studies exist to determine whether such therapy prolongs survival and improves quality of life in cats with CKD.

FLUID THERAPY

Summary of the evidence

Weak evidence (grade IV) exists to support long-term subcutaneous fluid therapy in cats with CKD.

Anecdotal evidence supports chronic administration of subcutaneous fluids to cats with advanced (stages 3–4) CKD that are prone to dehydration, or that respond to such therapy with a demonstrable improvement in clinical signs. Randomized controlled clinical trials are desirable to validate these recommendations and better identify those feline patients that will benefit most from this form of therapy.

Calcitriol therapy

The kidneys are responsible for converting 25-hydroxycholecalciferol to its most active metabolite, 1,25-dihydroxycholecalciferol, also known as calcitriol. Calcitriol is one of the major hormones responsible for calcium metabolism along with parathyroid hormone (PTH). Among calcitriol's important functions is modulation of PTH activity at the transcriptional level.¹⁹ As production of 1,25-dihydroxycholecalciferol is typically impaired in CKD, calcitriol deficiency may be one factor promoting renal secondary hyperparathyroidism. Parathyroid hormone has been proposed to act as a 'uremic toxin' and supplementing calcitriol may ameliorate a variety of supposed adverse effects of excess parathyroid hormone in patients with CKD. A study found that 8/26 cats with CKD had renal secondary hyperparathyroidism as detected by elevated serum PTH concentrations.²⁰

Use of calcitriol in patients with CKD is described in numerous textbooks and reviews.^{21–23} Investigators have reported that in cats with CKD receiving calcitriol therapy, patients: (1) were brighter and more alert, (2) had improved appetites, (3) were more physically active, and (4) lived longer.²¹ These findings were based on an uncontrolled survey of veterinarians who used calcitriol in their practice. A study of 10 cats with CKD showed that serum PTH concentrations were not significantly different after 14 days of calcitriol administration, regardless of whether calcitriol was administered daily or intermittently.²⁴ Adverse effects of calcitriol administration (specifically ionized hypercalcemia) were not observed in this short-term study.

However, many of the beneficial effects of calcitriol therapy in patients with CKD may be mediated through activation of vitamin D receptors on many tissues rather than modulation of PTH activity alone.²⁵ A 1-year randomized controlled clinical trial examined the effect of low-dose calcitriol therapy on progression of CKD and clinical signs in cats.²⁶ In this study, calcitriol was ineffective in altering renal mortality or improving appetite, activity or quality of life in cats with stages 2–4 CKD. In contrast, a 1-year randomized controlled clinical trial examining the effect of low-dose calcitriol therapy on progression of CKD and clinical signs in dogs demonstrated that calcitriol therapy was associated with a significant reduction in all-cause mortality compared with placebo.²

CALCITRIOL THERAPY

Summary of the evidence

Weak evidence (grade IV) from pathophysiological rationale, surveys of veterinarians and data from clinical trials in dogs and humans suggests a possible role for calcitriol in the management of CKD in cats. However, a short-term descriptive case series (grade III evidence) and a longer term unpublished, randomized controlled clinical trial (grade I evidence) failed to support the use of calcitriol in cats with CKD. Current evidence fails to support a recommendation either for or against use of calcitriol in cats with CKD.

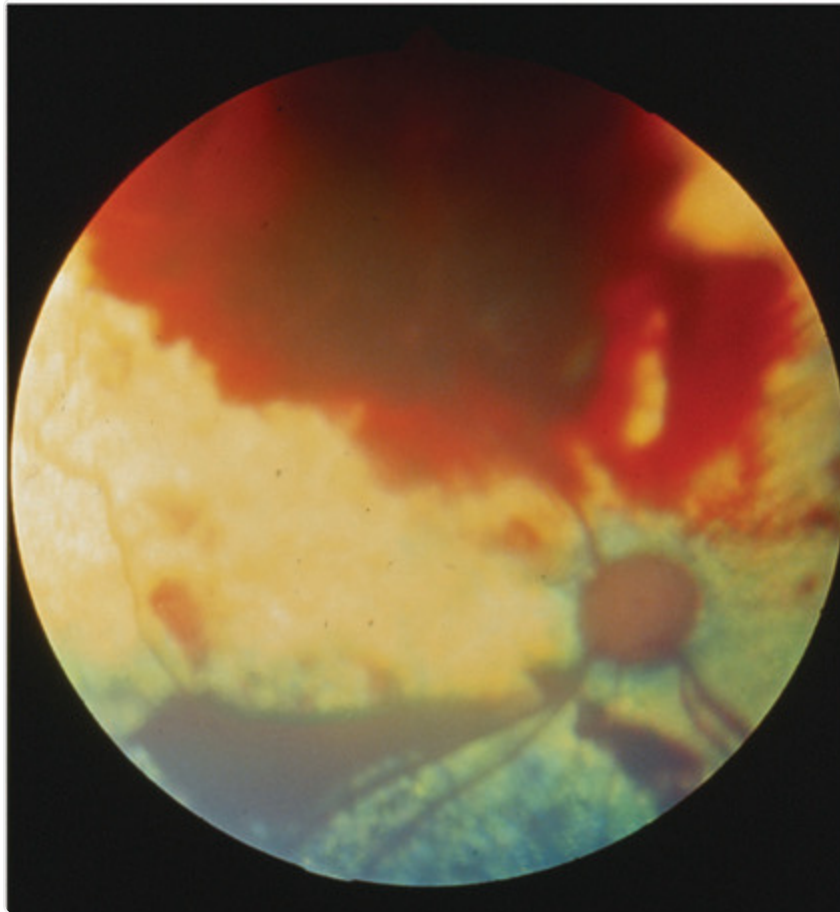
Antihypertensive therapy

Hypertension is a well recognized complication of CKD in cats. Mild to moderate systemic hypertension was originally reported in 17/28 cats that had spontaneous CKD.²⁷ More recent data suggest that the prevalence of hypertension in cats with CKD may be about 20%.²⁸

Sustained systemic hypertension can result in hypertensive retinopathy with retinal detachment, hemorrhage and blindness (Fig 1). Cats with such severe ocular manifestations probably reflect only a small percentage of pets with CKD and hypertension. More subtle ocular lesions such as retinal edema, retinal vessel tortuosity and subtle retinal hemorrhages may be much more common. Hypertension-related central nervous system disorders (eg, seizures, loss of balance, abrupt changes in personality, obtundation, etc) have also been observed.^{29–31} Management of

hypertension reportedly limits postoperative neurological disorders following renal transplantation in cats.^{[32](#)}

Fig 1.



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Retinal Image from a 10-year-old spayed female domestic shorthair cat with acute blindness. Chronic kidney disease and hypertension were diagnosed

Hypertension may also damage other organs including the kidneys, heart and blood vessels. A small study in cats with CKD suggested that glomerulosclerosis is more severe in hypertensive cats compared with normo-tensive cats.^{[33](#)}

Initiation of therapy for hypertension has been advocated for cats with clinical signs consistent with pressure-related end-organ damage (eg, hypertensive retinopathy with retinal detachment, acute-onset neurologic signs) and/or blood pressure values that persistently exceed 180/120 mmHg.^{4,34} However, the touted renoprotective benefit of antihypertensive therapy in cats is largely extrapolated from observations in humans and experimental studies in animals. The potential benefits of intervention in patients without clinical signs might include retarding progression of kidney disease, prolonging survival and reducing the incidence of hypertensive retinopathy and encephalopathy. In one study, proteinuria was the only variable significantly related to survival in hypertensive cats.⁸

Although specific evidence-based guidelines for treatment of systemic hypertension have not been definitively established for cats, studies have provided information on the effectiveness of some antihypertensive treatments. Currently, the calcium channel antagonist amlodipine besylate appears to be the drug of choice for managing hypertension in cats with CKD. Antihypertensive effects have been noted in uncontrolled clinical observations on the use of amlodipine in cats with spontaneous kidney disease.^{8,31,35,36} Blood pressure control with amlodipine appears to influence survival in hypertensive cats in a positive manner and lower blood pressures at the time of diagnosis indicate that treatment is more likely to be effective.⁸ Treatment with amlodipine can result in a significant reduction in proteinuria and, as mentioned, the level of proteinuria appears to be related to survival in hypertensive cats.⁸ Another study showed that amlodipine administration decreased systemic arterial blood pressure and dramatically reduced the prevalence of ocular lesions attributable to hypertension in cats with an induced model of renal failure.³⁷ In this last study, all cats were also fed a therapeutic renal food throughout the study period.

In contrast to amlodipine, angiotensin converting enzyme (ACE) inhibitors and β -blocking agents do not appear to be effective as sole agents for long-term management of systemic hypertension in cats.^{38,39} However, one study showed that the ACE inhibitor benazepril is well tolerated when administered with amlodipine to hypertensive cats and may assist in managing cats with poorly controlled blood pressure.⁴⁰ Amlodipine also appears to have intrinsic antioxidant activity.⁴¹ It is currently unknown whether this in vitro antioxidant activity results in significant improvement in endothelial function in patients with CKD or provides substantive clinical benefits.

ANTIHYPERTENSIVE THERAPY

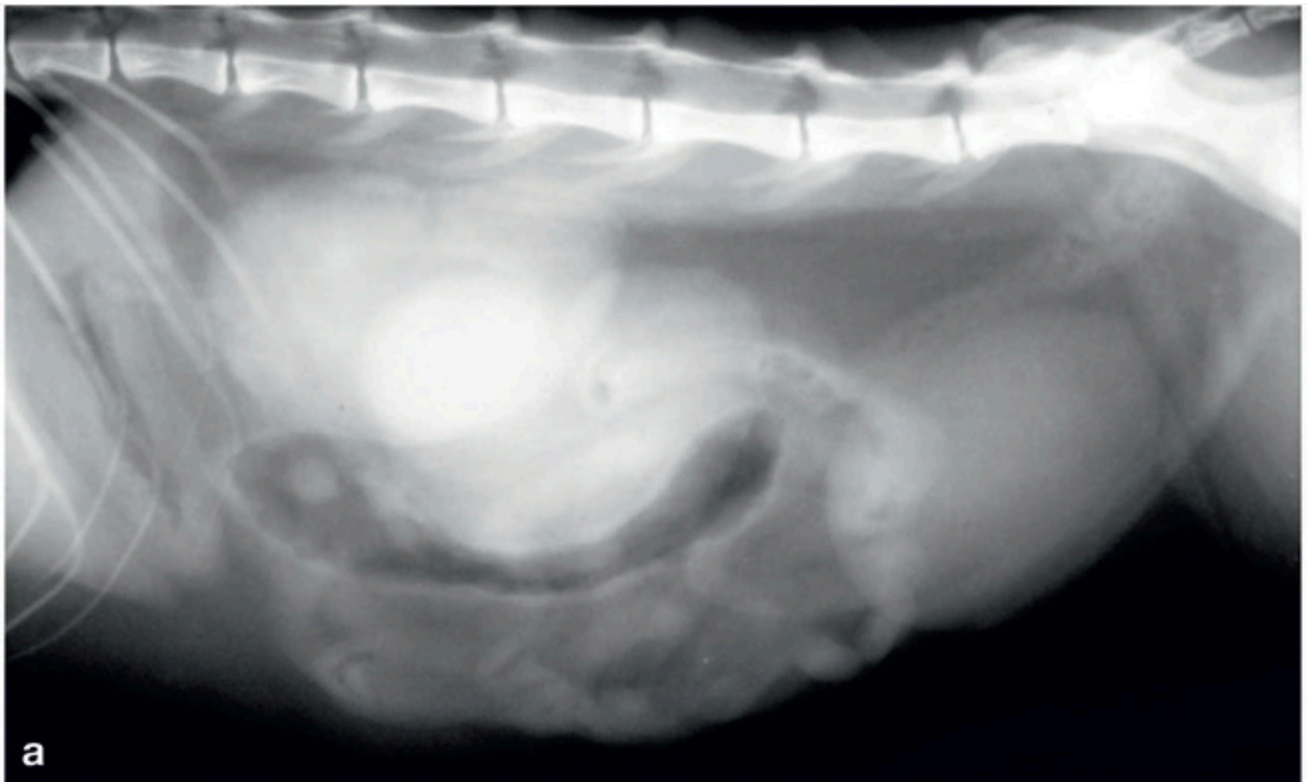
Summary of the evidence

There is moderate evidence (grade III) to recommend the use of amlodipine in the management of systemic hypertension in cats with CKD. Randomized controlled clinical trials are desirable to validate these recommendations and better identify those feline patients that will benefit most from this form of therapy.

ACE inhibitor therapy

Systemic and renal changes that are observed in patients with kidney disease may contribute to the progression of renal injury ([Figs 2](#) and [3](#)). Frequently implicated changes include increased systemic arterial pressure (systemic hypertension), increased intraglomerular pressure (glomerular hypertension), glomerular enlargement (glomerular hypertrophy) and altered glomerular perm-selectivity. Evidence suggests that therapy with ACE inhibitors preserves glomerular structure and function in patients with kidney disease. Administration of ACE inhibitors to rodents with reduced renal mass revealed that renoprotection is associated with lowering of glomerular capillary pressure, blood pressure and glomerular size.[42,43](#) Systematic reviews and meta-analysis of clinical studies of ACE inhibitor use in human patients have revealed beneficial effects in terms of slowing progression of non-diabetic kidney disease.[44,45](#) This beneficial effect seems to be mediated by factors in addition to decreasing blood pressure and urinary protein excretion and is greater in patients with proteinuria.[44,46](#)

Fig 2.



Lateral (a) and ventrodorsal (b) abdominal radiographs of a cat with chronic kidney disease and bilateral renomegaly. Lymphoma, polycystic kidney disease and feline infectious peritonitis were considered as potential causes. Diagnostic evaluation confirmed the presence of lymphoma

Fig 3.



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Kidneys from a cat that died as a result of acute progression of chronic kidney disease. Necropsy confirmed the presence of renal hypoplasia (small kidney) and renal hypertrophy (larger kidney)

The ACE inhibitor benazepril has been licensed in several countries for use in managing cats with CKD. [47](#) Two studies examined the physiological effects of benazepril in cats with induced kidney disease. Systemic arterial and glomerular capillary pressures were shown to be reduced and GFR was increased by such therapy. [48,49](#) However, the reduction in systemic blood pressure was small and a beneficial effect in reducing proteinuria was not evident. These initial studies failed to detect any evidence that administering benazepril resulted in long-term structural or functional renal protection. [50](#)

Data have recently been reported from randomized controlled clinical trials of benazepril use in cats with spontaneous CKD. [49,51-54](#) In two controlled studies, benazepril reduced proteinuria in cats with spontaneous CKD; however, it did not significantly prolong survival time. [51,54](#) Cats with a UPC > 1.0 had mean ($\pm$ standard deviation) survival times of 402 ± 202 days with benazepril ($n = 4$) compared with 149 ± 90 days with placebo ($n = 9$); however, this difference was not statistically significant. [51](#) Cats with initial UPC > 1.0 treated with benazepril had a significantly better appetite compared with cats treated with placebo. [51](#) While administration of ACE inhibitors lessens proteinuria in cats with CKD and proteinuria is a prognostic indicator for cats with CKD, it has yet to be established whether reduction of proteinuria with use of ACE inhibitors improves the long-term prognosis in cats with CKD. [50](#)

In a separate study, treatment with enalapril and benazepril did not change plasma renin activity, aldosterone concentration and indirect systolic arterial blood pressure in cats with hypertension associated with CKD. [39](#) However, another study showed that benazepril is well tolerated when administered with amlodipine in hypertensive cats and may assist in managing cats with poorly controlled blood pressure. [40](#)

ACE INHIBITOR THERAPY

Summary of the evidence

Although ACE inhibitors have been shown to reduce the magnitude of proteinuria in cats with spontaneous CKD (grade I evidence), weak evidence exists to support a recommendation for or against use of ACE inhibitors for the purpose of slowing progression and prolonging survival of cats with CKD (limited grade I and grade III evidence). In contrast, results of a randomized controlled clinical study provide moderate support for the recommendation of providing ACE inhibitors for the purpose of increasing appetite in cats with urine UPC > 1.0. Further randomized controlled clinical trials are desirable to validate these recommendations and better identify those feline patients that will benefit most from ACE inhibitor therapy.

Erythropoietic hormone replacement therapy

Low packed cell volume (PCV), erythrocyte count and hemoglobin are characteristic of cats with moderate to advanced CKD.^{[2,55](#)} The anemia is normocytic, normochromic, hypoproliferative and proportional to the severity of the kidney disease. The principal cause is hypoplasia of erythroid elements of the bone marrow secondary to inadequate production of erythropoietin by the kidneys. Shortened erythrocyte life span, erythropoietic inhibitor substances in plasma, chronic gastrointestinal blood loss, nutritional abnormalities (eg, iron deficiency) and bone marrow fibrosis may further exacerbate anemia in some patients with CKD. Recent studies in humans indicate that hypoxia from anemia of kidney disease may contribute to progression of CKD because the anemia reduces oxygen delivery within the kidney, further promoting hypoxia and progressive renal damage from oxidative stress.^{[56](#)}

Hormone replacement therapy using human recombinant erythropoietin (rHuEPO) has been shown to be effective in correcting anemia of CKD in cats. Uncontrolled clinical trials indicated a substantial improvement in appetite and quality of life associated with this treatment.^{[57](#)} Unfortunately, development of antibodies directed against rHuEPO has limited the effectiveness of this therapy in a substantial number of patients. As a consequence, it is best to select carefully those cases most likely to benefit from rHuEPO treatment. Generally, hematocrit or PCV guidelines

can be used to judge the severity of anemia; symptomatic cats with PCV < 20% are the best candidates.⁵⁵ In addition, the degree of azotemia, expected rate of progression of CKD, appetite and willingness to eat therapeutic renal foods, and rate of progression of anemia must all be considered in the risk-benefit analysis of when to start therapy.² Gastrointestinal blood losses may contribute to anemia in patients with CKD and should be investigated before making the decision to use rHuEPO therapy.

Darbepoietin- α is a longer acting form of rHuEPO that has been used in a limited number of cats with anemia of CKD. Compared with regular rHuEPO, darbepoietin has a longer half-life and greater potency enabling the same clinical efficacy with less frequent administration.⁵⁸ Although its use has not been supported by clinical studies in cats with CKD, anecdotal reports indicate that darbepoietin is similar in efficacy and safety to erythropoietin, with the possible benefit of a reduction in the incidence of neutralizing antibodies.

The efficacy and safety of recombinant feline erythropoietin therapy was recently evaluated in cats with anemia of CKD and those with CKD and red cell aplasia induced by rHuEPO therapy. Most cats demonstrated erythroid hyperplasia, reticulocytosis, increased hematocrit and improved quality of life (appetite and energy level) with treatment. Unexpectedly, some cats that initially responded to feline recombinant erythropoietin therapy subsequently developed anemia that was refractory to additional treatments.⁵⁹ Unfortunately, the recombinant feline erythropoietin used in this study is not currently available as a commercial product.

The effect of renal transplantation on erythropoiesis and serum erythropoietin concentrations in cats with CKD has been evaluated.⁶⁰ Although the erythropoietic response was highly variable in cats after renal transplantation, anemia resolved in most cats within 2 months after surgery, suggesting that hormone replacement therapy is not needed in this subset of cats.

ERYTHROPOIETIC HORMONE REPLACEMENT THERAPY

Summary of the evidence

Anemia is a serious complication in many feline patients with moderate to advanced CKD. There is moderate (grade III) evidence to support a recommendation to consider treatment of anemia of CKD using rHuEPO in carefully selected cases. Placebo-controlled clinical trials are needed to confirm the safety and efficacy of darbepoietin for managing anemia of CKD in cats. Future use of specific feline recombinant erythropoietin products may improve the overall success of therapy in cats with anemia of CKD should these products become available.

Potassium supplementation

An association between polyuric renal failure and hypokalemia has been recognized in cats by several investigators.^{[61](#)–[63](#)} Hypokalemia occurred in 19% of cases in a clinical series of cats with spontaneous kidney disease and was moderate to severe in more than half of these patients.^{[64](#)}

There is limited evidence that hypokalemia may be a cause of and contributing factor to CKD in cats, rather than simply a consequence of the disease. In an uncontrolled study, evidence of renal lesions and dysfunction developed in 3/9 cats fed a potassium-restricted, acidifying food for several months.^{[65](#)} However, it was not clear whether potassium depletion or hypokalemia preceded the onset of kidney disease. In another study, 4/7 cats with induced CKD fed a food containing 0.3% potassium (dry matter basis) developed hypokalemia but four cats with normal renal function fed the same food did not develop hypokalemia.^{[66](#)} Muscle potassium content has been shown to be decreased in normokalemic cats with spontaneous CKD, indicating that a total body deficit of potassium may develop well before the onset of hypokalemia.^{[67](#)} The latter findings support the concept that reduced renal function precedes the development of hypokalemia.

In many cats with CKD and hypokalemia, renal function may improve after potassium supplementation and restoration of normokalemia, suggesting that hypokalemia may be associated with a reversible, functional decline in GFR. Renal function was shown to be adversely affected in normal cats when an acidified, low-potassium food was fed.^{[68](#)} In that study, potassium depletion

and acidosis appeared to have additive effects in impairing renal function. Routine supplementation with low oral doses of potassium has been recommended for all cats with CKD. [62](#) This recommendation appears to be based on the hypothesis that, in some cats with CKD, hypokalemia and potassium depletion might promote a self-perpetuating cycle of declining renal function, metabolic acidosis and continuing potassium losses. It is proposed that potassium supplementation may stabilize renal function before potassium depletion exacerbates the disease.

The potential benefit of potassium supplementation has been examined; however, the clinical benefits of such therapy remain to be proven. [67,69](#) Although the value of providing supplemental potassium to cats with CKD and normokalemia has not been established, it is clear that muscle potassium and probably total body potassium stores may be reduced in at least some cats with CKD, presumably putting them at risk of hypokalemia. Feeding acidified foods or dietary acidifiers to cats with CKD may exacerbate their tendency to develop potassium depletion. [70](#) However, unpublished studies in cats with induced CKD failed to confirm an adverse effect of mild to moderate metabolic acidosis on hypokalemia or renal lesions. [71](#) There also appears to be an inverse relationship between serum potassium concentrations and blood pressure in cats. [72](#)

POTASSIUM SUPPLEMENTATION

Summary of the evidence

Although supported only by weak (grade III) evidence, potassium supplementation is generally advocated for cats with CKD and hypokalemia, even in the absence of overt clinical signs. There is weak (grade IV) evidence to support a recommendation either for or against routine supplementation of potassium for all cats with CKD for preventing hypertension or progression of CKD. Given that current feline renal foods are generally supplemented with additional potassium, randomized controlled clinical trials are required to establish whether potassium supplementation beyond the amount found in current feline renal foods is necessary or beneficial.

Antioxidant supplementation

The kidney accounts for 10% of the whole body oxygen consumption and is a site of significant aerobic metabolism that produces reactive oxygen species as by-products.⁷³ Increased free radical production and antioxidant depletion have been reported in human patients with CKD.⁷⁴ Increased free radicals appear to cause renal injury and increased oxidative stress may contribute to the progression of CKD.⁷⁵⁻⁷⁷

In a study, cats with spontaneous CKD were fed a dry food supplemented with the anti-oxidants vitamins E and C, and 3-carotene for 4 weeks. When compared with healthy cats, cats with stage 2 CKD had a tendency to develop oxidative stress.⁷⁸ The antioxidant supplements significantly reduced DNA damage in cats with CKD as evidenced by reduced serum 8-hydroxy-2'-deoxyguanosine (8-OHdG) concentrations, and comet assay parameters of DNA damage. Serum 8-OHdG concentrations generally reflect oxidation of DNA and comet assay shows the breakdown of DNA structure in the cells measured.

ANTIOXIDANT SUPPLEMENTATION

Summary of the evidence

Cats with early spontaneous CKD show a tendency towards enhanced oxidative stress, which may worsen as CKD progresses. Dietary supplementation with antioxidant nutrients such as vitamin E, vitamin C and α -carotene significantly reduced oxidative damage to DNA in cats with spontaneous CKD. Based on limited evidence, cats with CKD might benefit from antioxidant fortification of therapeutic renal foods.

Randomized controlled clinical studies are needed to confirm the potential benefits of dietary antioxidant supplementation on progression of CKD and important clinical outcomes (survival, quality of life).

Alkalinization therapy

Metabolic acidosis appears to be a common complication of CKD in cats,^{61,64} and alkalinization therapy designed to correct metabolic acidosis may be an important part of the overall management of such patients. A cross-sectional study involving 59 cats with CKD showed that over half of patients with severe disease had acidemia and low plasma bicarbonate concentrations.^{79,80} These data also suggested that biochemical evidence of severe metabolic

acidosis does not generally occur in cats until late in the course of CKD.⁷⁹ As discussed earlier, there is also a putative relationship between hypokalemia, potassium depletion, chronic metabolic acidosis and CKD in cats.⁸¹

Alkalinization therapy has been recommended for patients with moderate to severe metabolic acidosis associated with CKD based on pathophysiological justification and extrapolation from findings in other species. Potential benefits of alkalinization therapy in patients with CKD include:

- improving clinical signs that may be caused by uremic acidosis;
- preventing the catabolic effects of metabolic acidosis on protein metabolism, thereby
- promoting adaptation to dietary protein restriction;
- limiting skeletal changes resulting from bone buffering; and
- modifying progression of kidney disease.^{2,81,82}

Chronic metabolic acidosis appears to impose an unnecessary metabolic risk that can easily be corrected by administration of potassium citrate or sodium bicarbonate when acidosis is appropriately confirmed.

In an investigation of cats with induced CKD, an acidifying (AD) or non-acidifying (NAD) diet was fed for 39 weeks.⁷¹ Hyperchloremic metabolic acidosis developed in the AD group (mean venous pH = 7.27, compared with 7.33 in the NAD group); the net acid excretions in this group were six-fold higher than in the NAD group. Cats with induced-CKD fed the AD for 6 months did not develop progressive glomerular dysfunction or renal tubulointerstitial injury when compared with cats fed a NAD. This suggests that levels of dietary acidification typically used in manufactured adult maintenance foods may not promote progression of feline CKD. A double-masked placebo-controlled cross-over study examined the effect of supplementing the diet with potassium gluconate (2 mEq twice daily) for 12 weeks.^{69,83} This regimen had no effect on venous blood pH or bicarbonate ion concentrations although it did result in an increase in urine pH. No other controlled clinical trials have been performed to confirm that appropriate alkalinization therapy will significantly alter the progression of disease or clinical signs in cats with spontaneous CKD. Most therapeutic renal foods used in the management of CKD have increased dietary buffering capacity and are already less acidifying than many standard feline maintenance foods.

ALKALINIZATION THERAPY

Summary of the evidence

There is weak (grade IV) evidence to support therapy of acidemia using alkalinizing agents such as sodium bicarbonate or potassium citrate for the purposes of slowing progression of CKD and minimizing clinical signs or uremic acidosis. The benefit of alkali supplementation over and above that provided by therapeutic renal foods in cats with CKD remains to be established. Randomized controlled clinical trials are necessary to validate recommendations for routine use of this therapy, especially in the light of the fact that oral supplementation with alkalinizing agents is a treatment that is often not well accepted by the patient and must be given multiple times per day for the remainder of the pet's life.

Dietary phosphorus restriction and intestinal phosphate binders

Phosphorus is retained in CKD, eventually resulting in hyperphosphatemia, which in turn promotes renal secondary hyperparathyroidism. The role of dietary phosphorus intake and hyperphosphatemia in pets with spontaneous CKD was first discussed nearly 30 years ago.^{[84,85](#)} Today, there appears to be a consensus of opinion that phosphate retention is a major cause of progression of CKD in many species.^{[9](#)} In human beings with CKD receiving hemodialysis therapy, the adjusted relative risk of mortality was stable in patients with serum phosphate concentrations below 6.5 mg/dl, but increased significantly above this level.^{[86](#)} Similar data for mortality risk associated with hyperphosphatemia in cats with spontaneous CKD are lacking. Nonetheless, treatment targets for serum phosphorus concentrations have been established for cats with CKD: serum phosphorus concentration is recommended to be maintained below 4.5 mg/dl (1.45 mmol/l), 5.0 mg/dl (1.6 mmol/l) and 6.0 mg/dl (1.9 mmol/l) in cats with stages 2, 3 and 4 CKD, respectively.^{[9](#)}

Hyperphosphatemia appears to be a reliable clinical indicator of hyperparathyroidism in cats with CKD.^{[87,88](#)} Hyperphosphatemia is detected in approximately 60% of cats with CKD, with the prevalence increasing as renal function declines. In a study of cats with advanced CKD, the prevalence of renal secondary hyperparathyroidism was reported to be 84%.^{[88](#)} In this study, 87%

of cats with clinical signs and 47% of clinically normal cats with only biochemical evidence of CKD were diagnosed with renal secondary hyperparathyroidism. Hyperparathyroidism was even detected in nine cats with CKD having normal serum calcium and phosphorus concentrations.

In a model of induced kidney disease in cats, high dietary phosphorus intake (1.56% dry matter) for up to 11 months was associated with renal mineralization, fibrosis and mononuclear cell infiltration.⁸⁹ Low levels of phosphorus intake (0.42% dry matter) were not associated with these histological abnormalities. Progressive renal dysfunction was not detected in either group during the timeframe of the study. In at least some cats with spontaneous CKD, dietary management with low-phosphorus feline renal foods alone may normalize hyperparathyroidism.⁸⁸

Intestinal phosphate binding agents may be useful in further reducing phosphate retention and hyperparathyroidism in cats with CKD when nutritional management alone (therapeutic renal foods) is insufficient in achieving serum phosphorus concentration targets. However, experimental or controlled clinical studies establishing the effectiveness of adding intestinal phosphate binding agents to dietary phosphate restriction in cats with CKD are limited. Clinical reports and anecdotal evidence suggest that Intestinal phosphate-binding agents are useful in reducing serum phosphate concentrations, especially in uremic, hyperphosphatemic animals.^{9,22} Lanthanum carbonate octahydrate has recently been developed as an intestinal phosphate binding agent and studied in azotemic, normophosphatemic cats with a model of CKD.⁹⁰ Addition of this agent to a moist high-phosphorus (2.2% phosphorus, dry matter basis) feline maintenance food fed to cats with induced CKD resulted in reduced phosphorus digestibility and increased phosphorus excretion.⁹⁰ Controlled clinical studies of this and other phosphate binders to confirm effectiveness in slowing progression of spontaneous CKD in cats have not been reported.

Intestinal phosphate binders are not well tolerated by some cats and may impair food intake. These agents must be given with every meal to be effective, which may decrease owner acceptance and usefulness of this therapy.

DIETARY PHOSPHORUS RESTRICTION AND INTESTINAL PHOSPHATE BINDERS

Summary of the evidence

While there is good evidence for use of dietary phosphorus restriction in therapeutic renal foods for management of CKD in cats (see below), there is only weak evidence (grade IV) supporting the recommended treatment targets for serum phosphorus concentrations. Consequently, only weak evidence exists for adding intestinal phosphate binders to dietary phosphorus restriction for additional control of hyperphosphatemia and hyperparathyroidism. This is relevant in the light of the fact that treatment with phosphate binders is often not well accepted by the patient and must be given multiple times per day for the remainder of the pet's life. The value of strict, target-based recommendations for management of serum phosphorus concentrations should be confirmed by randomized controlled clinical trials.

Therapeutic renal foods

Therapeutic foods have been used for over 50 years in veterinary patients with CKD. Compared with typical commercial and homemade pet foods, therapeutic renal foods usually have reduced protein, phosphorus and sodium, and increased dietary buffering capacity, soluble fiber, B-complex vitamins, antioxidants, potassium and omega-3 fatty acids.⁸¹ Veterinarians are often confronted with the decision of whether to recommend switching to a therapeutic renal food or to continue the current diet with the view that eating any food is better than risking reduced food intake by attempting a potentially unwanted dietary change. This clinical challenge underlines the importance of assessing the quality and strength of evidence underlying the use of therapeutic renal foods.

Several published clinical trials have provided evidence in support of feeding therapeutic renal foods when managing cats with stages 2-A CKD. A retrospective study compared the survival times of cats newly diagnosed with CKD that had been fed a normal diet or one of seven different types of therapeutic feline renal foods.⁹¹ In this study, the median survival time of 175 cats with CKD that received conventional foods was 7 months, whereas the median survival time of 146 cats with CKD fed one of the therapeutic renal foods was 16 months.⁹¹ A prospective study comparing feeding a

renal food with not making a dietary change, showed a striking enhancement of survival time associated with feeding the renal food.⁹² The control group in this study was composed of cats that refused to eat the renal food and continued their usual diet. A principal limitation of this study is that cats electing not to consume the renal food may have had an intrinsically worse prognosis unrelated to the food consumed. However, the magnitude of the difference in outcome (median survival time was increased nearly 2.5 times when the renal food was fed) suggests that the clinical benefit of feeding the renal food was likely to be real.

Of all the treatments recommended for cats with CKD, the only one that has been shown (in a randomized controlled clinical study) to decrease uremic episodes and mortality is therapeutic renal food.

The single most important thing you can do to increase acceptance of therapeutic renal foods is gradually transition, sometimes over a period of at least 3–4 weeks.

The effectiveness of a therapeutic renal food in cats with stages 2 and 3 CKD was evaluated by a randomized controlled clinical trial.⁹³ This study examined whether clinically important benefits accrued when cats with CKD consumed a renal food compared with a standard feline maintenance food. Other than being randomly assigned to either the renal food or the maintenance food, cats were managed in an identical manner with respect to other treatment interventions. Cats fed the maintenance food had a significantly greater number of uremic episodes (26%) compared with cats fed the renal food (0%). In addition, a significant reduction in renal-related mortality was observed in cats fed the renal food (0% versus 22%). Importantly, significant adverse effects of feeding the renal food were not detected in the study and cats fed the renal food maintained body weight and body condition during the 2-year study period.

Seemingly, the greatest problem with advocating renal foods for cats with CKD has been acceptance of the food. In most instances this can be overcome by carefully correcting metabolic complications of CKD and gradually introducing the renal food over several weeks (see box, page 206). In a recent clinical trial, most cats with CKD (90%) readily accepted and continued eating a renal therapeutic food using a gradual transition period and were no more likely to refuse the renal therapeutic food than a more typical adult maintenance food.⁹³

THERAPEUTIC RENAL FOODS

Summary of the evidence

There is strong evidence supporting the recommendation that cats with stages 2, 3 and 4 CKD be fed a therapeutic renal food (grade I evidence for stages 2 and 3 CKD; grade III evidence for stage 4 CKD) for the purpose of minimizing uremic episodes and reducing premature mortality.

Assisted feeding

Malnutrition is usually detected as weight loss, declining serum albumin or total plasma protein concentrations, anemia and evidence of decreased muscle mass. In patients with CKD, malnutrition usually results from inadequate food intake. Commercially available foods designed for patients with CKD contain sufficient protein, calories and other nutrients to sustain adequate nutrient intake when consumed in appropriate quantities. Some cats with stages 3 and 4 CKD fail to eat sufficient food voluntarily, regardless of palatability or nutrient content.^{[94](#)}

When malnutrition is evident or suspected, clinical recommendations often include a stepwise approach designed to facilitate adequate food intake.^{[2,95](#)} The first step is to ensure that metabolic and other medical causes of decreased appetite have been corrected including dehydration, gastrointestinal hemorrhage, metabolic acidosis, hypokalemia, anemia, urinary tract infection and drug-associated anorexia. When metabolic and other medical causes of anorexia have been excluded or corrected, therapy for uremic gastroenteritis should be initiated. A study has shown that hypergastrinemia is common in cats with naturally developing CKD and may contribute to gastric hyperacidity, uremic gastritis, gastrointestinal tract bleeding and associated clinical signs of anorexia and/or vomiting.^{[96](#)} Therapy for gastrointestinal complications of uremia usually includes administration of an H₂ antagonist such as famotidine, which may be combined with an anti-emetic and a gastric mucosal protectant such as sucralfate.^{[97](#)}

If therapy for uremic gastroenteritis fails to restore normal appetite, assisted feeding should be considered. Long-term use of esophagostomy or percutaneous gastrostomy tubes has been successful for delivering food, extra water and medications to patients with CKD. Anecdotal

reports suggest that tube feeding can reverse the progressive weight loss associated with CKD and patients can have extended periods of improved quality of life.[2,55,95](#)

ASSISTED FEEDING

Summary of the evidence

Although there is only weak (grade IV) evidence supporting the use of assisted feeding in the management of CKD in cats, there is no proven effective alternative. Clinical observations suggest that assisted feeding may reverse malnutrition and improve quality of life in many patients. Randomized controlled clinical trials would assist in validating the benefits and risks of assisted feeding in patients with CKD and better identify those cats that would benefit from this form of therapy.

10 tips to encourage acceptance of therapeutic renal foods

1 Provide owner education

Educate pet owners about the role of nutritional management in prolonging survival time and quality of life in cats with kidney disease. In order for treatment to have the best chance of success, owners must commit valuable resources (ie, their time and money), which is more likely to occur if they understand the benefits of their efforts.

2 Implement early

Begin nutritional management sooner rather than later. Current evidence supports feeding a renal food when serum creatinine is ≥ 2 mg/dl ($176\mu\text{mol/l}$). Waiting until creatinine is higher or the patient is showing uremic signs (eg, inappetence, vomiting) is not advised. Uremic patients are less likely to accept a new food and therefore will not receive the optimal benefits of a renal therapeutic food.

3 Transition gradually

The single most important thing you can do to increase patient acceptance of a therapeutic renal food is gradually transition to the new food. The transition period should be a minimum of 7 days; however, many cats may need a transition of 3–4 weeks or longer. It is critical to discuss the need for this transition with pet owners, otherwise they are likely to buy the new food, go home, and switch from the old food to the new food at the next meal. In this scenario, many cats will refuse to eat the new food, resulting in an unhappy owner and a patient that will not receive the benefits of the prescribed nutritional management.

4 Provide old and new

One option for transitioning to a renal food is to mix the old and new food, gradually adding more of the new food over time. Another approach is to

provide both foods (old and new) in food dishes side by side. This latter technique assists with gradual transition and also allows cats to express their preferences. For more information see The Indoor Cat Initiative website (www.vet.osu.edu/indoorcat )

5 Use a suitable food dish

When transitioning cats from dry to moist food, use a flat food dish (eg, saucer) rather than a bowl. This avoids the cat's whiskers rubbing on the food dish, which could affect acceptance of a new food.

6 Avoid stressful periods

Avoid offering therapeutic renal foods in stressful periods (eg, to sick and/or hospitalized cats, or during force-feeding); a food aversion may develop causing decreased acceptance of the food when the patient is feeling better. Put another way — while hospitalized, do not feed your feline patients the food you want them to eat for the rest of their lives. One option would be to feed a maintenance food that avoids excessive protein, phosphorus and sodium until the patient is feeling better and then begin gradual transitioning to a therapeutic renal food.

7 Temperature may be important ...

Use fresh food at room temperature. Some patients may eat refrigerated food that is warmed, but others will only eat food from a newly opened container. Some patients may eat food that has been refrigerated and stored in a plastic container (versus being stored in the original can).

8 ... So may texture and formulation

Offer foods with different textures (eg, minced formulas) or formulations (dry versus moist). Some pets may have shown a preference for either dry or moist food when healthy but, when they develop kidney disease, their preferences may switch (eg, a cat that has eaten only dry food all its life may only eat moist food once kidney disease occurs).

9 Add flavor enhancers

Add flavor enhancers (low-sodium chicken broth, tuna juice, oregano, brewer's yeast, small amount of regular food) to enhance palatability and encourage the patient to eat all the therapeutic food. Excessive use of other food may decrease the beneficial effects of nutritional management.

10 If all else fails, try a different brand

If, having followed all the above steps, there is still reluctance to eat a therapeutic renal food, switch to a different brand. While commercially available renal foods have general features in common, they are not the same foods. In addition, individual pets may express a preference for one brand over another. Avoid giving the owner samples of several different brands of foods at once; this could result in the cat developing a food aversion to all foods, especially if owners offer each sample at successive meals or on consecutive days.

Dialysis

In human medicine, hemodialysis is the most common means of providing replacement or supplemental renal excretory function. Hemodialysis has been used successfully to manage clinical signs of uremia in cats,^{[94,98-100](#)} Although hemodialysis is primarily indicated for managing cats with potentially reversible acute renal failure, there may be patients with CKD for which hemodialysis should be considered as a suitable adjunct to other forms of therapy, specifically in preparing cats for possible renal transplantation. Hemodialysis is most likely to be beneficial when serum urea nitrogen concentrations exceed 90 mg/dl (32 mmol/l urea) and serum creatinine concentrations exceed 8 mg/dl (700 μ mol/l).^{[98,99](#)}

Recent investigations have focused on recruiting the gastrointestinal tract as a means of excreting wastes normally eliminated by the kidney; in essence, enteric dialysis. Polymers designed for oral administration adsorb uremic toxins and excess fluid in the gastrointestinal tract for excretion in the feces.^{[24](#)} These polymers have therapeutic potential as another extra-renal clearance mechanism to be used in conjunction with hemodialysis or other forms of therapy in patients with CKD. Further controlled clinical studies with oral polymer therapy are anticipated. In addition, a

probiotic product purported to enhance gastrointestinal excretion of nitrogenous waste products has recently entered the veterinary market. The manufacturer proposes that enhanced intestinal clearance of these nitrogenous wastes will ameliorate clinical signs of uremia.

Renal transplantation

Renal allograft transplantation has become a viable therapeutic option for cats with CKD. Long-term immunosuppression with a combination of oral corticosteroids and ciclosporin has proven successful in cats, although regular monitoring is required.^{[101,102](#)} However, severe and often fatal complications of renal transplantation may occur in cats.^{[32,103-108](#)}

Recommended criteria for selection of feline transplant recipients include:

- early decompensated CKD for which conventional therapy is no longer effective;
- weight loss of no greater than 20% of healthy body weight;
- no history of recent urinary tract infection;
- no concurrent serious medical conditions (eg, inflammatory bowel disease);
- no evidence of cardiac dysfunction;
- negative tests for chronic systemic viral infections.^{[109](#)}

Other criteria to evaluate when considering renal transplantation include the emotional and financial commitment by the owner to immediate and long-term care of the animal, as well as access to a veterinarian familiar with managing transplant recipients should complications arise.

Table 1.

Summary of evidence grades supporting recommendations for therapy of feline CKD

Evidence grade	Therapeutic recommendations
Grade I	Some therapeutic renal foods ACE inhibitors (reducing magnitude of proteinuria; increasing appetite in cats with UPC > 1.0)
Grade III	Some therapeutic renal foods Antihypertensive therapy (amlodipine in cats with CKD and proven hypertension) Recombinant human or feline erythropoietin (suitably anemic cats with CKD) Potassium supplementation (cats with CKD and hypokalemia) Dietary phosphorus restriction (cats with stages 3 and 4 CKD)
Grade IV	Fluid therapy (long-term administration of subcutaneous fluids) Potassium supplementation (all cats with CKD) Intestinal phosphate binders Alkalinizing therapy (acidemic cats with CKD) Assisted feeding (anorexia and malnutrition in cats with CKD) ACE inhibitors (non-proteinuric CKD) CKD — chronic kidney disease, ACE = angiotensin converting enzyme

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DIALYSIS

Summary of the evidence

In general, it is probably inappropriate to recommend hemodialysis for long-term therapy of cats with CKD until studies have been performed confirming the clinical effectiveness of such therapy, including consideration of the effect of such therapy on overall quality of life for the cat and owner. Preliminary evidence in species other than cats provides some experimental basis for the concept of 'enteric dialysis'. Randomized controlled clinical trials are needed to confirm the effectiveness of the veterinary-licensed probiotic in minimizing clinical signs of uremia in cats with CKD.

RENAL TRANSPLANTATION

Summary of the evidence

Weak evidence currently exists to support a recommendation either for or against renal transplantation as an alternative to medical management of cats with CKD. Studies comparing clinical outcomes in cats with comparable CKD treated with medical management and renal transplantation are desired. High cost, technical complications, inconvenience and limited availability will continue to preclude the routine use of renal transplantation for many cats with CKD.

KEY POINTS

The concepts of evidence-based medicine can be readily applied to management of CKD in cats.

For management of CKD in cats, the highest quality and strength of evidence exists for use of specific therapeutic renal foods ([Table 1](#)). This is the only evidence-supported therapeutic intervention that should be recommended routinely for initial management of CKD in cats.

Most cats will accept a dietary change if it is started early in the course of disease and if a gradual transition period is used.

Moderate strength and quality of evidence exists for use of antihypertensive agents (patients with hypertension associated with CKD), ACE inhibitors (proteinuric kidney disease), erythropoietic hormone replacement (patients with anemia) and renal transplantation.

At present, the lowest strength and quality of evidence exists for use of subcutaneous fluid therapy, calcitriol, alkalinizing agents, intestinal phosphate binders, assisted feeding and dialysis.

Randomized controlled clinical trials are needed to validate the benefits and risks of many treatments recommended and used in patients with CKD to better identify those patients that would benefit most from other forms of management.

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